

Equivalent representation of rational functions.

$$f(x) = \frac{p(x)}{q(x)}$$

↓ Long Division

$$\begin{array}{r} q \\ q(x) \overline{) p(x)} \\ \hline r \end{array}$$

$$f(x) = q + \frac{r}{q(x)}$$

Rational functions & zeros.

$f(x) = \frac{p(x)}{q(x)} \Rightarrow$ this $f(x)$ has zeros/x-intercepts
for x such that $p(x)=0$ & $q(x) \neq 0$

$\frac{p(x)}{q(x)}$ we want x values
that make $p(x)=0$
& $q(x) \neq 0$.

if $q(x)=0$, the function becomes undefined.

Rational Functions & vertical asymptotes.

$f(x) = \frac{p(x)}{q(x)}$, we have a vertical
asymptote for x -values where $q(x)=0$.



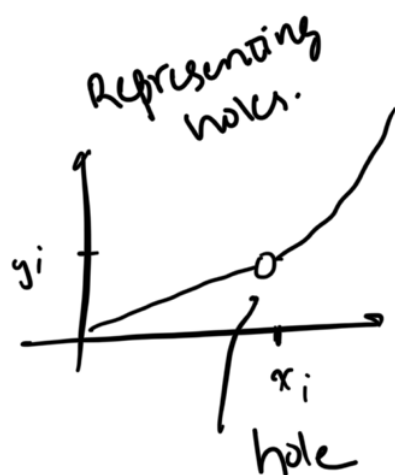
$$x_i \rightarrow Q(x_i) = 0$$

$$\lim_{x \rightarrow x_i} f(x) = \infty$$

Rational Functions & holes.

A function $\frac{P(x)}{Q(x)}$ has removable discontinuity called holes if the "multiplicity" of a common factor in the numerator is higher than or equal to that in the denominator.

$$f(x) = \frac{x^2 + 9x + 8}{x + 8} \rightarrow \frac{(x+8)(x+1)}{(x+8)}$$



$(x+8)$ is the common factor.
 its multiplicity in the numerator is equal to that in the denominator. $\Rightarrow$ we have a hole.

Summary of finding x & y co-ordinates of holes.

① common factor will provide the x-coordinate of holes

$$\therefore (x+8) \rightarrow x = -8 \text{ (x-coordinate)}$$

②

Plugging in x-coordinate of hole to
"reduced" $f(x)$ to get y-coordinate.

$$\frac{\cancel{(x+8)}(x+1)}{\cancel{(x+8)}} = (x+1) \leftarrow \begin{array}{l} \text{of hole} \\ \text{plug } x = -8 \end{array}$$

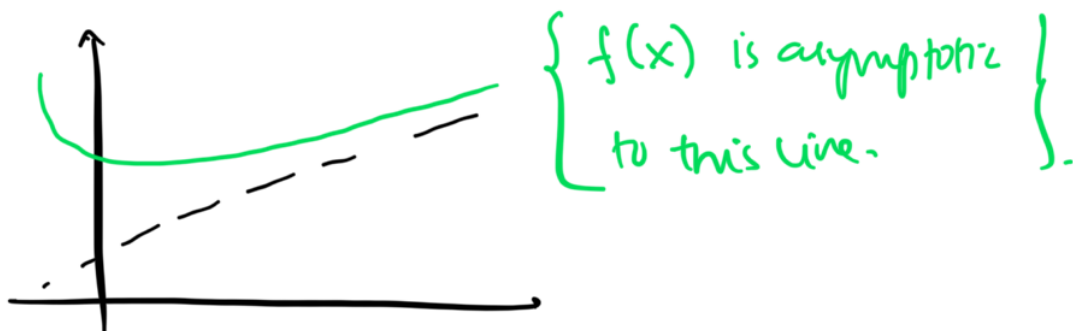
$$y = -8 + 1 \\ = -7$$

hole exists at $(-8, -7)$.

3.5 Rational Functions & End Behavior.

→ this tells us how $f(x)$ would
look like when $x \rightarrow \infty$

We had some rules for this, which
you should refer to.



if $f(x) = \frac{10x^4 - 2x^2 + 3x + 7}{\dots}$, evaluate the

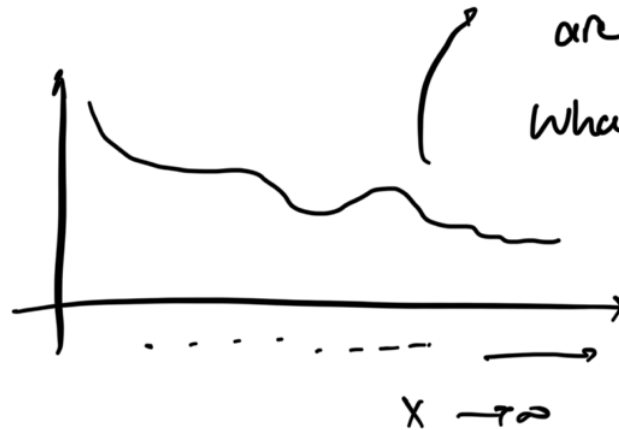
$$5x^4 + x^3 - 2$$

following

(a) $\lim_{x \rightarrow -\infty} f(x)$

(b) $\lim_{x \rightarrow +\infty} f(x)$

What does this limit ask about?
End Behavior of the function. As $x \rightarrow \infty$, we



are looking at the End behavior.
What are we looking at as $x \rightarrow \infty$?

$$f(x) = \frac{10x^4 - 2x^2 + 3x + 7}{5x^4 + x^3 - 2}$$

} degree n
} degree m

$$a_n x^n + a_{n-1} x^{n-1} \dots a_1 x + a_0$$

$$b_m x^m + b_{m-1} x^{m-1} \dots b_1 x + b_0$$

$$\lim_{x \rightarrow +\infty} f(x)$$

$$n = 4$$

$$m = 4$$

We have a horizontal asymptote at

$$y = \frac{a_n}{b_m}$$

End Behavior for $f(x)$.

which tells us $y =$

$$\frac{(10x^4) - 2x^2 + 3x + 7}{(5x^4) + x^3 - 2} = \frac{a_n x^n + a_{n-1} x^{n-1} \dots a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} \dots b_1 x + b_0}$$

what are a_n & b_m ?

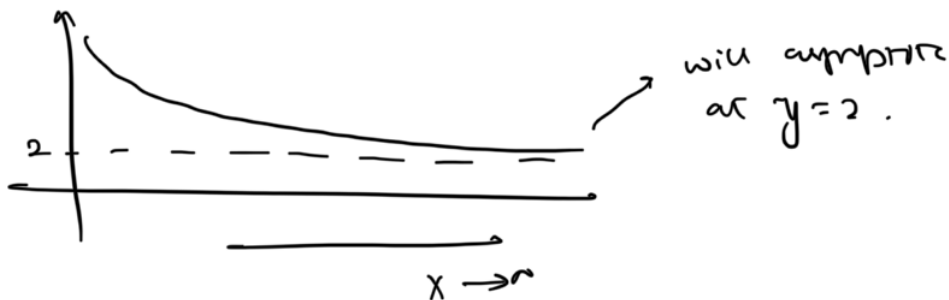
$a_n x^n = 10x^4$, $b_m x^m = 5x^4$

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$$a_4 = 10, \quad b_4 = 5$$

$$y = \frac{a_n}{b_m}, \text{ since } \frac{n}{m} = 4 \Rightarrow y = \frac{a_4}{b_4}$$

$$y = \frac{10}{5} = 2.$$



Graphing Rational Functions

1. Factor the numerator & denominator of the rational function (and find & remove the holes).
2. Find all possible x and y-intercepts
3. Determine the degree of the numerator & denominator $\Rightarrow$ $\lim_{x \rightarrow \infty} f(x)$ determine asymptotic behavior
4. Find all the vertical asymptote $\frac{p(x)}{q(x)}$ all $x \Rightarrow q(x) = 0$
5. Find any symmetry in the graph by evaluating $f(-x)$ i.e. $f(-x) = f(x)$
 $f(-x) = -f(x)$
6. The vertical asymptotes will divide

the graph into regions in every region graph at least one point to determine whether the graph goes up or down.



7. sketch the graph using all of this information.

Graph $f(x) = \frac{x^2 - 64}{x^2 - 16}$

(1) Factor the Numerator &

Denominator. $\therefore a^2 - b^2$

$$f(x) = \frac{(x+8)(x-8)}{(x+4)(x-4)}$$

= (a+b)(a-b)
factored.
it

(2) Find all possible x & y-intercepts

↙ x-intercepts

x-intercepts $x = 8, x = -8$

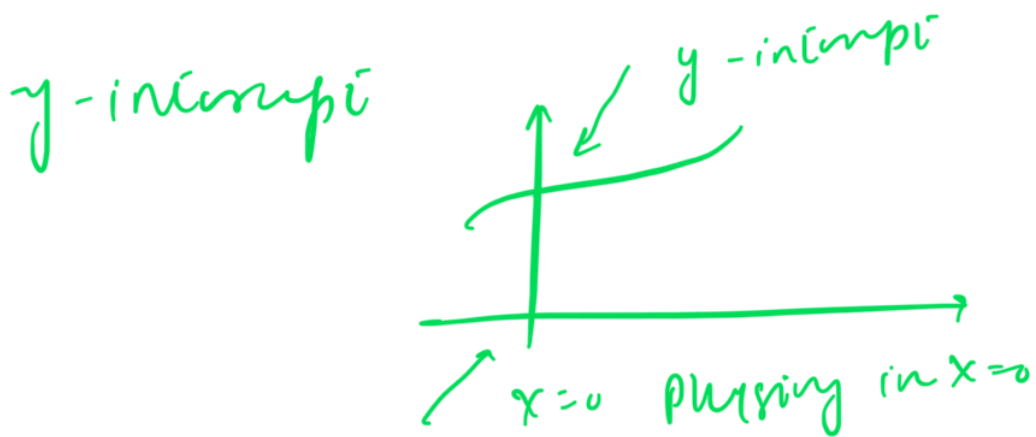
Don't

make denominator zero!

then neither make $f(x)$ undefined.

For example $f(x=4) \rightarrow$ This wouldn't be accepted.
 $f(4) = \frac{(4+8)(4-8)}{(4+4)(4-4)} \rightarrow 0$ undefined.

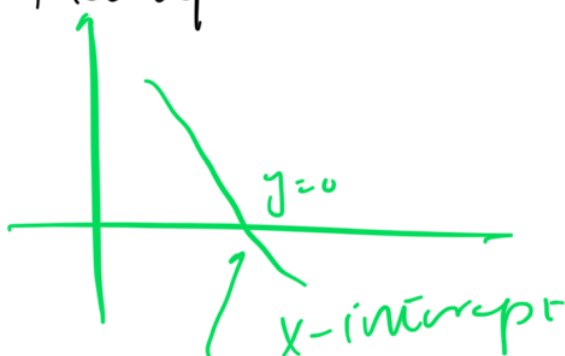
$$= \frac{(12)(-4)}{(8) \times 0} = \frac{(12)(-4)}{0} \rightarrow \text{undefined}$$



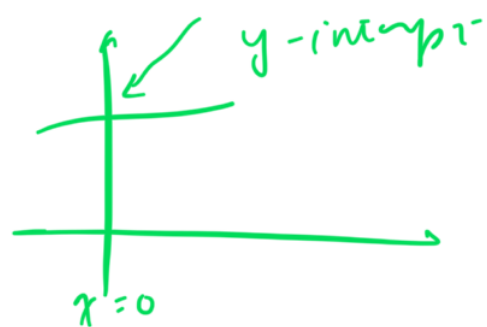
$$f(0) = \frac{(0+8)(0-8)}{(0+4)(0-4)} = \frac{(8)(-8)}{(4)(-4)}$$

x=0 at the y-intercept

$$= \frac{-64}{-16} = 64/16 = 8/2 = 4$$



x-intercepts: -8, 8



y-intercept: 4

3. Find the end behavior

$$f(x) = \frac{x^2 - 64}{x^2 - 16} \rightarrow \boxed{n=2} \quad \boxed{m=2} \quad \left[\text{Degrees} \right]$$

$$\text{form (b)} \Rightarrow \underline{\underline{n=m}}$$

$$y = \frac{a_n}{b_m}$$

horizontal asymptote



$$\boxed{y = \frac{1}{1}}$$

Actual polynomial

$$\rightarrow (1)x^2 - 64$$

general polynomial form $\{ a_2 x^2 + a_1 x + a_0 \}$
 your plot will asymptote to $y = 1$



$$\boxed{\lim_{x \rightarrow \infty} f(x) = 1}$$

4. Find all vertical asymptotes.

$$f(x) = \frac{(x-8)(x+8)}{(x+4)(x-4)}$$

$$\Rightarrow \boxed{(x+4)(x-4)} \quad Q(x)$$

$$\boxed{Q(x) = 0}$$

vertical asymptotes

Roots $\Rightarrow (x+4)(x-4) = 0$
for
the $\rightarrow$ these are x -values that make
denomi. $f(x)$ undefined.
-naïve

$$f(0) = \frac{(-8)(8)}{(4)(-4)}$$

$$f(4) = \frac{(4-8)(4+8)}{(4+4)(4-4)} \quad \text{undefined.}$$

$$f(-4) = \frac{\quad}{(-4+4)(-4-4)} \quad \text{undefined.}$$

We have vertical asymptotes

at $x=4, x=-4$.

Step-3 Evaluate $f(-x)$

to check for symmetry
i.e. would the function
be Even or Odd?

$$f(x) = \frac{x^2 - 64}{x^2 - 16}$$

$$\begin{aligned} f(-x) &= \frac{(-x)^2 - 64}{(-x)^2 - 16} \\ &= \frac{x^2 - 64}{x^2 - 16} \end{aligned}$$

$$f(x) = f(-x)$$

$\Rightarrow$ makes $f(x)$ an even
function.

AN Example Even function

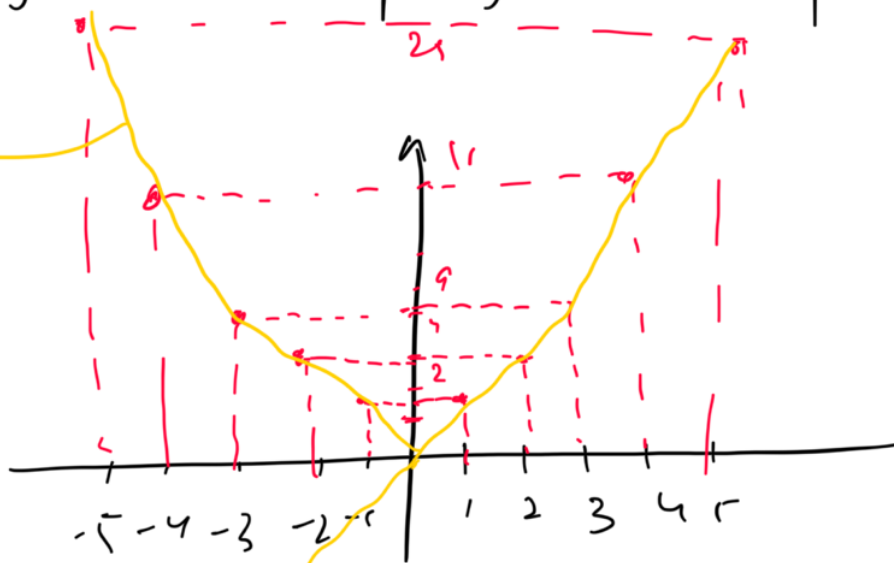
if a function $f(x)$ is even then $[f(x) = f(-x)]$

$$\begin{array}{c|c} f(1) = 2 & f(2) = 4 \\ f(-1) = 2 & f(-2) = 4 \end{array}$$

$$\begin{array}{c|c|c} f(3) = 9 & f(4) = 16 & f(5) = 25 \\ f(-3) = 9 & f(-4) = 16 & f(-5) = 25 \end{array}$$

Even

function.



odd function

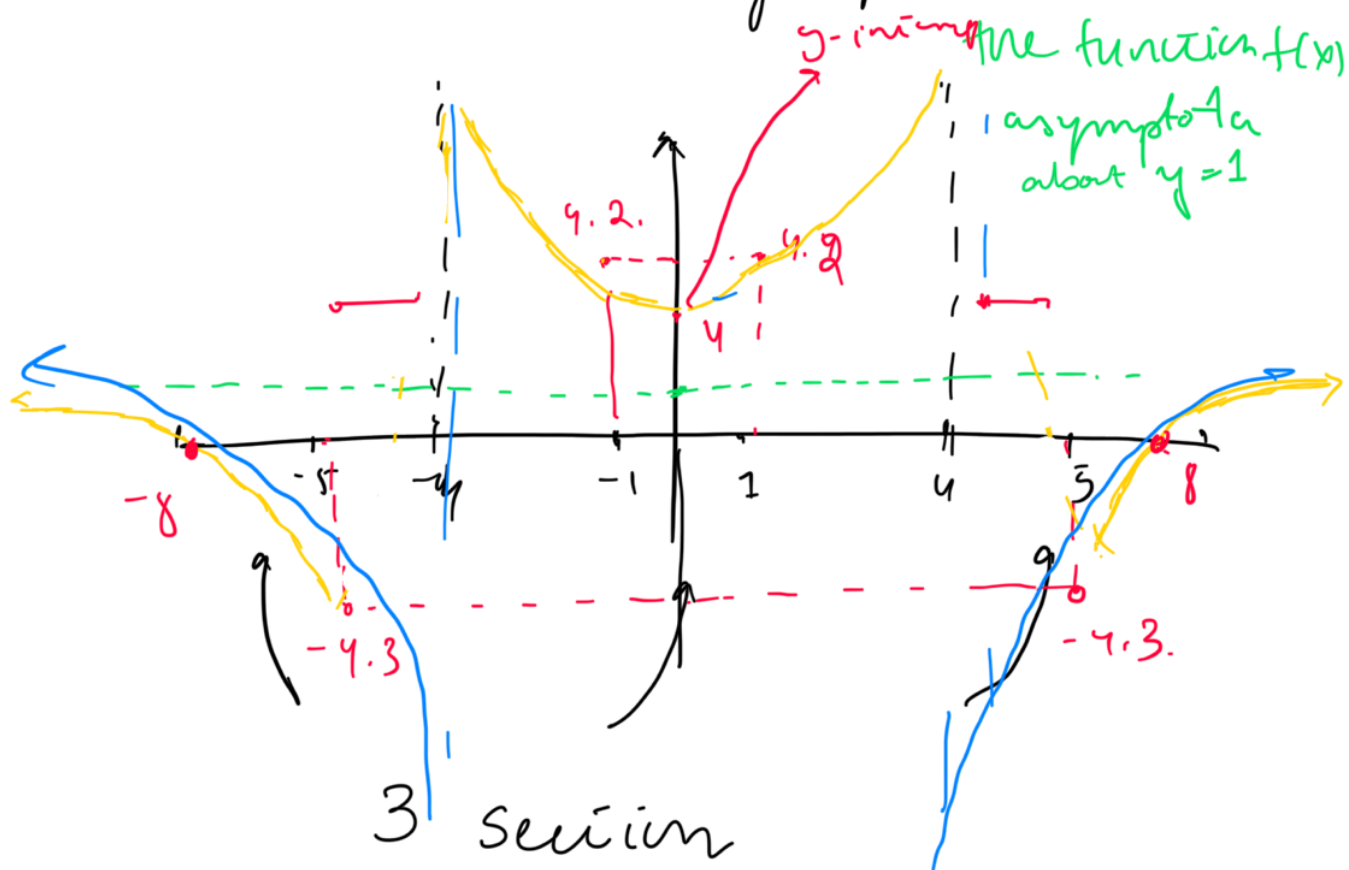
$$f(-x) = -f(x)$$

“

A FUNCTION DOESN'T
HAVE TO BE EVEN OR
ODD - IT CAN BE

NEITHER. 99

Step-6 Figure out the behavior of the function between the vertical asymptotes!



we need to evaluate our function in these 3 sections, to understand its behavior.

$$f(-5) = \frac{(-5)^2 - 64}{(-5)^2 - 16} = -\frac{39}{9} = -4.3$$

$$f(1) = \frac{1 - 64}{1 - 16} = \frac{-63}{-15} = 4.2$$

$$f(0) = \frac{-64}{-16} = \frac{64}{16} = 4$$

Example - 2 Plotting Rational Functions

$$g(x) = \frac{(x+1)(3x^2-12x+2)}{(x+1)(x-4)}$$

Step-1 Factorize the Rational function

$$3x^2 - 12x + 2$$

Using the quadratic formula.

$$a=3, b=-12, c=2$$

$$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(3)(2)}}{2(3)}$$

$$x = \frac{12 \pm \sqrt{144 - 24}}{6}$$

$$x = \frac{12 \pm \sqrt{120}}{6} = \frac{12 \pm 2\sqrt{30}}{6}$$

$$x = \frac{6 \pm \sqrt{30}}{2}$$

holes

$$g(x) = \frac{(x+1)(3x^2 - 12x + 2)}{(x+1)(x-4)}$$

common
factor

if there is a repeating factor, & the multiplicity of that factor in the numerator $\geq$ multiplicity in the denominator, you will have holes.

the mult. of $(x+1)$ in

the numerator is the same as its mult. in the denominator $\Rightarrow$ you have holes

Where is the hole?

$$f(x) = \frac{(\cancel{x+1})(3x^2 - 12x + 2)}{(\cancel{x+1})(x-4)}$$

$$g(x) = \frac{(3x^2 - 12x + 2)}{(x-4)}$$

the x -coordinate for the hole is at $(x+1)=0$

$$x = -1$$

$$g(-1) = \frac{3(-1)^2 - 12(-1) + 2}{(-1-4)}$$

$$= \frac{3(1) + 12 + 2}{-5}$$

$$= -5$$

$$= -3.4$$

hole is located at $\boxed{(-1, -3.4)}$

Step-2 Find all the
x & y-intercepts

x-intercept $(0, 0.174)$

$$\frac{6 \pm \sqrt{30}}{2}, (0, 3.825)$$

Revis from
here.

$$g(x) = \frac{(x+1)(3x^2 - 12x + 2)}{(x+1)(x-4)}$$

you want take $x = -1$ as a root / x-inter

$$(x+1)(3x^2 - 12x + 2) = 0$$

y-intercept

$$g(0) = \frac{(0+1)(3(0) - 12(0) + 2)}{(0+1)(0-4)}$$

$$= \frac{(1)(0 - 0 + 2)}{-4}$$

$$= \cancel{1/-4} = \left(-\frac{1}{2} \right)$$

↗ y-intercept

Step-3 Determine End Behavior

- Numerator num degree $n = 3$.
- Denominator num degree $m = 2$.

$n > m \Rightarrow$ the 3rd case

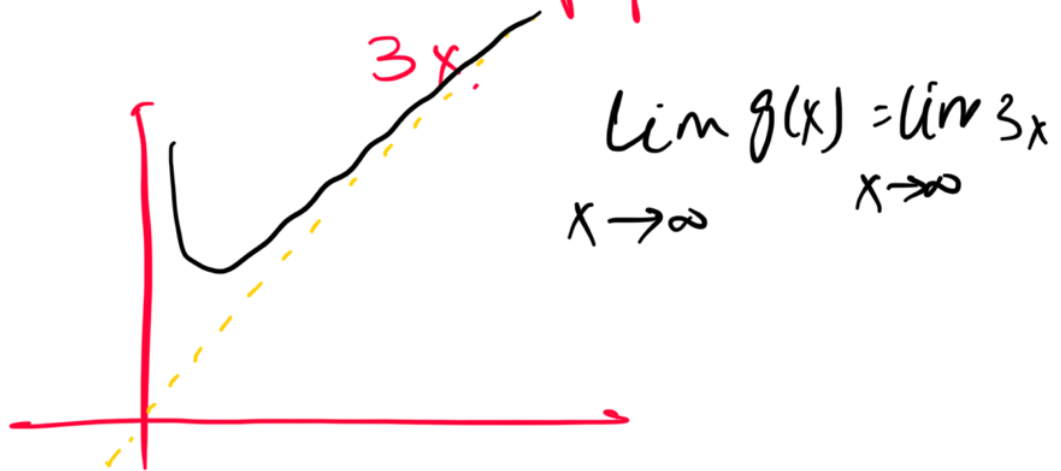
Long-Division

$$f(x) = \frac{(\cancel{x+1})(3x^2 - 12x + 2)}{(\cancel{x+1})(x-4)}$$

$$\begin{array}{r} 3x \\ x-4 \overline{) 3x^2 - 12x + 2} \\ \underline{(3x^2 - 12x)} \end{array}$$

$$g(x) = \frac{3x}{2} + \frac{2}{(x-4)}$$

our final plot
will asymptote about



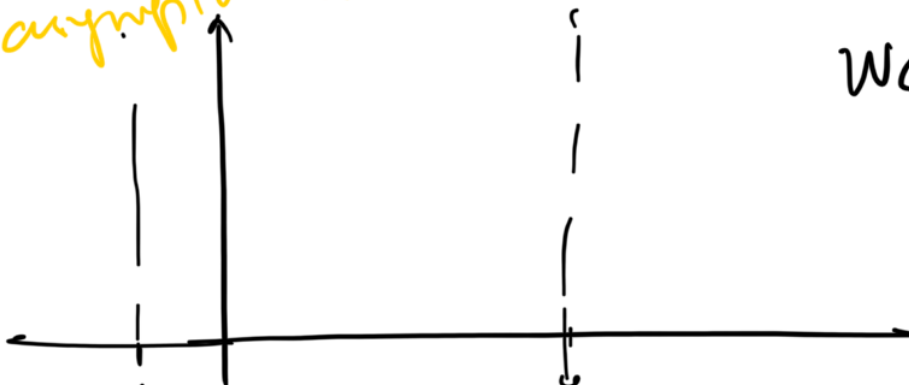
Step-4 Find all the vertical asymptotes

they will be at

hole &

not a
vertical
asymptote

$x = -1, x = 4$
& we removed it in step-1



We will
again have

3 sections.

Step - 5 Evaluate

$x = -1$ $x = 4$

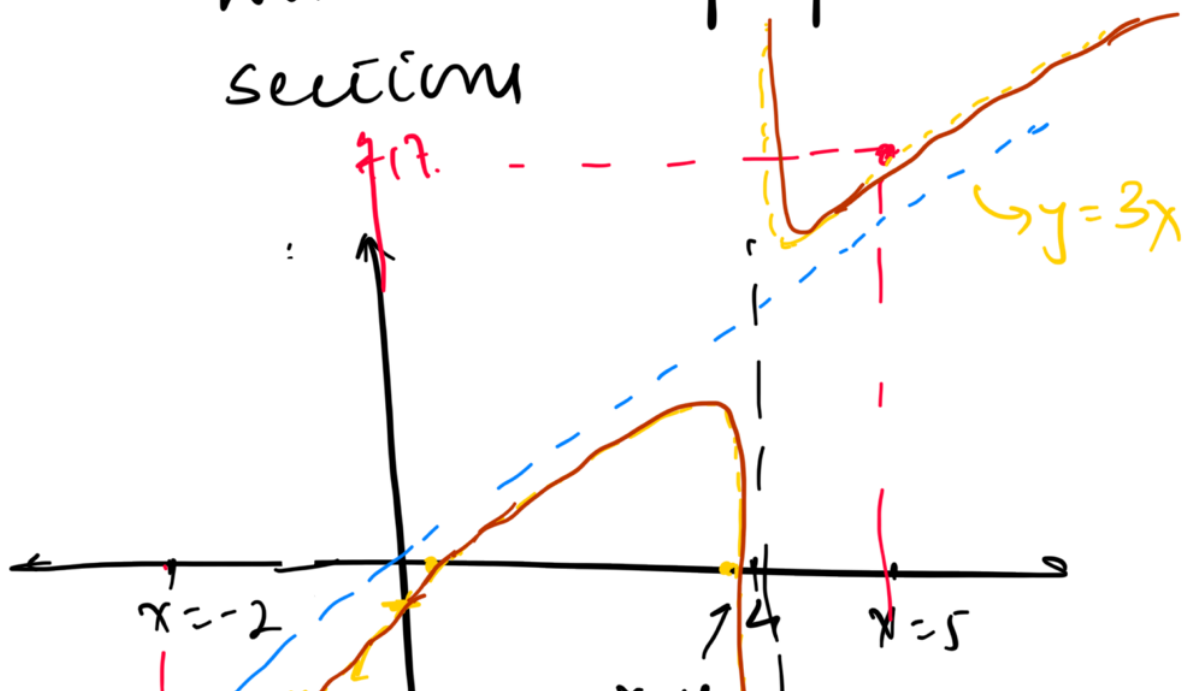
we know $x = -1$ as it is when we have a hole rather than vertical asymptote

$f(-x)$ will not have any symmetry. i.e.

it's neither even or odd

$f(-x) \neq f(x)$
 $f(-x) \neq -f(x)$

Step - 6 Examine behavior within the asymptote sections





$$g(5) = 17$$

$$g(-2) = -6.333$$

x-intercepts

$$(0.1742, 0)$$

$$(3.8257, 0)$$